New q-ary Sequential Locally Recoverable Codes from Product Codes

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Notation

- $[n] := \{1, 2, \ldots, n\}$.
- An $[n, k]_q$ code is a linear code defined over $\mathbb{F}_q$ with length n and dimension k.
- An $[n, k, d]_q$ code is an $[n, k]_q$ code with minimum distance d.
- $\mathcal{C}^{*\mathcal{S}}$ is the puncturing of a length n code $\mathcal{C}$ at indices $\mathcal{S} \subset [n]$.

Locally Recoverable Codes (LRCs)

Definition

- An $[n, k]_q$ linear code $\mathcal{C}$ is said to have **locality** r if any symbol can be recovered from at most r other symbols of the code (independent of choice of codeword).
- We call these sets of (at most) r symbols the **$(r, \mathcal{C})$-recovery sets** of index $i \in [n]$.
- An $[n, k]_q$ code with locality r (preferably with $r < k$) is called an **(n, k, r)-locally recoverable code (LRC)**.

Definition

- For $i \in [n]$, let $a(i)$ be the number of $(r, \mathcal{C})$-recovery sets for i.
- The **repair alternativity** of $\mathcal{C}$ is defined to be $a = \min \{a(i)\}_{i\in[n]}$.

Parallel and Sequential LRCs

Definition

An $[n,k]_q$ linear code $\mathcal{C}$ is said to be an **(n,k,r,t)-parallel locally recoverable code (PLRC)** if for each $E \subseteq [n]$ of size $|E| \leq t$, each $i \in E$ has a $(r,\mathcal{C})$-recovery set $R_i \in [n] \setminus E$.

Definition

An $[n,k]_q$ linear code $\mathcal{C}$ is said to be an **(n,k,r,t)-sequential locally recoverable code (SLRC)** if for each $E \subseteq [n]$ of size $|E| \leq t$, there is an ordering of $E = \{i_1, \ldots, i_t\}$ such that each $i_j \in E$ has an $(r,\mathcal{C})$-recovery set $R_{i_j} \subseteq [n] \setminus \{i_j, i_{j+1}, \ldots, i_t\}$.

Proposition ([Son+18])

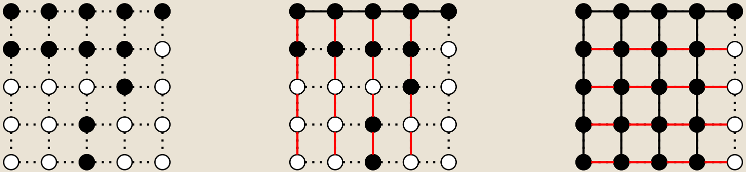
$\mathcal{C}$ is an (n,k,r,t)-SLRC if and only if for any nonempty $E \subseteq [n]$ of size $|E| \leq t$, there exists an $i \in E$ such that i has a recovery set $R_i \subseteq [n] \setminus E$.

Example SLRC

Example

Consider the following as a $(25, 4, 2, 15)$-SLRC where:

- black nodes are unerased symbols,
- white nodes are erased symbols,
- lines are 2-recovery sets;



(a) Initial pattern (b) First iteration (c) Second iteration

Products of Codes

Definition

Let $A = (a_{ij})_{i,j}$ be an $m_1 \times n_1$ matrix and B be an $m_2 \times n_2$ matrix. Then the **Kronecker product** $A \otimes B$ is the $m_1 m_2 \times n_1 n_2$ matrix:

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n_1}B \\ \vdots & \ddots & \vdots \\ a_{m_1 1}B & \cdots & a_{m_1 n_1}B \end{bmatrix}.$$

Definition

Let $\mathcal{A}$ and $\mathcal{B}$ (respectively) be $[n_1, k_1, d_1]_q$ and $[n_2, k_2, d_2]_q$ linear codes with generator matrices $G_{\mathcal{A}}$ and $G_{\mathcal{B}}$. The **product code** $\mathcal{A} \otimes \mathcal{B}$ is the $[n_1 \cdot n_2, k_1 \cdot k_2, d_1 \cdot d_2]_q$ linear code with generator matrix $G_{\mathcal{A}} \otimes G_{\mathcal{B}}$.

Two Code Construction

Theorem

For $i \in \{1,2\}$, let $\mathcal{C}_i$ be an (n_i, k_i, r_i, t_i)-SLRC over $\mathbb{F}_q$ with repair alternativity a_i and let

$$N = n_1 n_2, \qquad K = k_1 k_2, \qquad T = t_1 t_2 + t_1 + t_2,$$
$$r = \max\{r_1, r_2\}, \qquad A = a_1 + a_2.$$

Then $\mathcal{C}_1 \otimes \mathcal{C}_2$ is a q-ary (N, K, r, T)-SLRC with repair alternativity $\geq A$ and rate $\frac{K}{N}$.

Moreover, if $t_i + 1$ is equal to the minimum distance of $\mathcal{C}_i$ (for $i = 1, 2$), then $T + 1$ is equal to the minimum distance of $\mathcal{C}_1 \otimes \mathcal{C}_2$, i.e., the local recovery capacity stays maximal under the code product.

General Construction

Corollary

For all $i \in [\ell]$, let $\mathcal{C}_i$ be an $(n_i, k_i, r_i, t_i)-$SLRC over $\mathbb{F}_q$ with repair alternativity a_i and let

$$N_\ell = \prod_{i=1}^{\ell} n_i, \qquad K_\ell = \prod_{i=1}^{\ell} k_i, \qquad T_\ell = \left(\prod_{i=1}^{\ell}(t_i + 1)\right) - 1,$$

$$r_\ell = \max_{i \in [\ell]}\{r_i\}, \qquad A_\ell = \sum_{i=1}^{\ell} a_i.$$

Then $\mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_\ell$ is a q-ary $(N_\ell, K_\ell, r_\ell, T_\ell)$-SLRC with repair alternativity $A \geq A_\ell$ and rate $\frac{K_\ell}{N_\ell}$.

Moreover, if $t_i + 1 = \mathrm{dist}(\mathcal{C}_i)$ for all $i \in [\ell]$, then $T + 1 = \mathrm{dist}(\mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_\ell)$.

Recovery Capability

Theorem

Let $\mathcal{C}$ be an $[n, k, d]_q$ code with $1 \leq k < n$, let $\ell \geq 2$, and let μ denote the number of erasures. Then the ℓ-fold product $\mathcal{C}^{\otimes \ell} := \mathcal{C} \otimes \cdots \otimes \mathcal{C}$ has the following erasure recovery behavior:

- *If $\mu \leq \ell(d-1)$, we can recover everything with only one copy of the $[n, k, d]_q$ code (in parallel).*
- *If $\ell(d-1) < \mu < d^\ell$, we can fully sequentially recover, but we cannot necessarily recover in parallel.*
- *If $d^\ell \leq \mu \leq \min(n^\ell - k^\ell, \ell n^{\ell-1}(d-1))$, we can possibly recover depending on the erasure pattern.*
- *If $\min(n^\ell - k^\ell, \ell n^{\ell-1}(d-1)) < \mu$, we cannot fully recover.*

Explicit Constructions from MDS and BCH Codes

For $q = 3$ and $q = 5$, we explicitly checked the parameters up to $k = 10$ of all products of:

- $\mathcal{P}_q$: $[q+1, 2, q]_q$ MDS codes,
- $\mathcal{D}_{q,n}$: $[n, n-1, 2]_q$ MDS codes,
- $\mathcal{R}_{q,n,k}$: $[n, k, n-k+1]_q$ (possibly extended) Reed-Solomon codes,
- $\mathcal{B}_{q,n,d}$: BCH codes of length n with designed distance d over $\mathbb{F}_q$,
- $\mathcal{C}^{*\mathcal{S}}$: puncturings of any of the above at indices $\mathcal{S}$.

Explicit Constructions: $q = 3$, $k \in \{4, 6\}$, $n < 50$

q	k	r	ℓ	Code	t	$a \geq$	Construction
3	4	2	2	$[16, 4, 9]_3$	8	6	$\mathcal{P}_3 \otimes \mathcal{P}_3$
				$[12, 4, 6]_3$	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,3}$
				$[9, 4, 4]_3$	3	2	$\mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$
3	6	2	2	$[32, 6, 15]_3$	14	6	$\mathcal{P}_3 \otimes \mathcal{B}_{3,8,5}$
				$[24, 6, 10]_3$	9	4	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,5}$
3	6	3	2	$[16, 6, 6]_3$	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,4}$
				$[12, 6, 4]_3$	3	2	$\mathcal{D}_{3,4} \otimes \mathcal{D}_{3,3}$

Explicit Constructions: $q = 3$, $k = 8$, $n < 50$

q	k	r	ℓ	Code	t	$a \geq$	Construction
3	8	4	2	$[20, 8, 6]_3$	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,5}$
				$[15, 8, 4]_3$	3	2	$\mathcal{D}_{3,5} \otimes \mathcal{D}_{3,3}$
3	8	3	2	$[32, 8, 12]_3$	11	8	$\mathcal{P}_3 \otimes \mathcal{B}_{3,8,4}$
				$[28, 8, 9]_3$	8	5	$\mathcal{P}_3 \otimes \mathcal{B}^{*\{1\}}_{3,8,4}$
				$[24, 8, 8]_3$	7	6	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,4}$
				$[21, 8, 6]_3$	5	3	$\mathcal{D}_{3,3} \otimes \mathcal{B}^{*\{1\}}_{3,8,4}$
				$[18, 8, 4]_3$	3	3	$\mathcal{D}_{3,3} \otimes \mathcal{B}^{*\{1,5\}}_{3,8,4}$
3	8	2	3	$[48, 8, 18]_3$	17	7	$\mathcal{P}_3 \otimes \mathcal{P}_3 \otimes \mathcal{D}_{3,3}$
				$[36, 8, 12]_3$	11	5	$\mathcal{P}_3 \otimes \mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$
				$[27, 8, 8]_3$	7	3	$\mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$

Rate Comparisons to Other SLRCS with $r = 2$

t	3	5	7	8	9	11
this work	**0.44**	**0.33**	**0.30**	0.25	**0.25**	**0.22**
$\frac{r}{r+t}$ [WZL15]	0.40	0.29	0.22	0.20	0.18	0.15
1st const. [Son+18]	**0.44**	**0.33**	**0.30**	**0.26**	0.24	**0.22**
$\frac{r}{2r+t-1}$ [Son+18]	0.33	0.25	0.20	0.18	0.17	0.14

Our constructions used here are all over $\mathbb{F}_3$ and for fixed r and t generally achieve the rate with shorter length than the others, e.g., for $t = 5$, ours $(\mathcal{P}_3 \otimes \mathcal{D}_{3,3})$ has length $n = 12$ vs. $n = 24$ for the first construction of [Son+18].

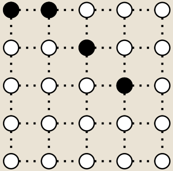
Our construction's notable restriction is that $t + 1$ cannot be prime.

Worst-case Iteration Counts

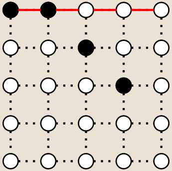
If we allow all "parallel" copies of codes to be recovered simultaneously, what is the worst-case scenario for number of iterations if we can fully recover?

Example

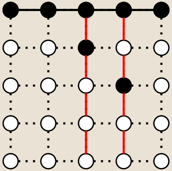
Again, consider the $(25, 4, 2, 15)$-SLRC from before (but with different erasures) where:



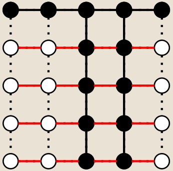
(a) Initial pattern



(b) First iteration



(c) Second iteration



(d) Third iteration

Worst-case Iteration Counts: Known Values

Let $I(n, \ell, r)$ be the worst case number of iterations for any recoverable error pattern on an ℓ-fold copy of a length n code with locality r. Then

- $I(n, 1, r) = 1$ for all n and all r (trivial; just a single line),
- $I(n, 2, r) \leq 2r - 1$ for all n and all r,
- $I(3, 3, 2) = 7$ (by brute force checking),
- $I(4, 3, 2) = 10$ (by brute force checking),
- $I(3, 4, 2) \geq 14$ (by pathological example).

Resolvable Configurations

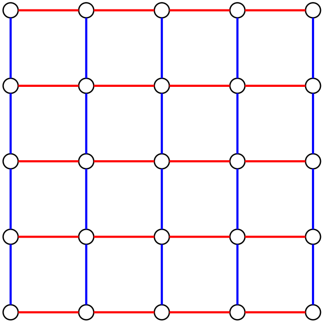
Definition ([Gév19])

Let $\mathcal{P}$ be a set of **points** and $\mathcal{B}$ be a set of **blocks** i.e. subsets of P such that:

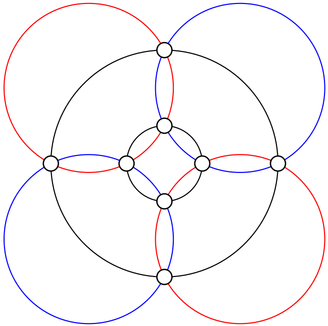
- $|\mathcal{P}| = v$,
- $|\mathcal{B}| = b$,
- each point is contained in r blocks,
- each block contains k points,
- the blocks can be partitioned into r **parallel classes** where the blocks of each class partition P.

Then $(\mathcal{P}, \mathcal{B})$ is said to be a $\boldsymbol{(v_r, b_k)}$**-resolvable configuration**.

Examples of Resolvable Configurations



(a) A $(25_2, 10_5)$-resolvable configuration



(b) A $(8_3, 6_4)$-resolvable configuration (the Miquel configuration)

Conclusion

Open questions:

- Can we build good SLRCs from products of other classes of codes?
- Algorithm for finding optimal sequences for recovery.
- Conjectures regarding iteration counting:
  - $I(n, 3, 2) = 10$ for all $n \geq 4$,
  - if $\ell \geq 4$ or if $r \geq 3$ and $\ell \geq 3$, there is a value of n such that $I(n, \ell, r)$ can be arbitrarily large (by specific placement of erasures).
- Consider iteration counting with regards to other resolvable configurations.

Thanks for your attention

Bibliography

[Gév19] G. Gévay. "Resolvable configurations". In: *Discrete Applied Mathematics* 266 (2019), pp. 319–330.

[Son+18] W. Song, K. Cai, C. Yuen, K. Cai, and G. Han. "On Sequential Locally Repairable Codes". In: *IEEE Transactions on Information Theory* 64.5 (2018), pp. 3513–3527. DOI: 10.1109/TIT.2017.2711611.

[WZL15] A. Wang, Z. Zhang, and M. Liu. "Achieving arbitrary locality and availability in binary codes". In: *2015 IEEE International Symposium on Information Theory (ISIT)*. 2015, pp. 1866–1870.